A Hybrid NLP Model for Fake News Detection on the WELFake Dataset

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Abstract

Automatic detection of fake news in digital media remains a significant challenge due to the volume of online content, which exceeds the capacity for manual fact-checking. This paper presents a hybrid natural language processing classification system that combines three feature representations to distinguish fake from real news articles on the WELFake dataset, a benchmark corpus of 72,134 articles merged from four sources (Verma et al., 2021).

The proposed system uses three parallel branches. The first branch applies TF-IDF vectorization with 50,000 unigram and bigram features. The second branch uses Stanford GloVe 100-dimensional pretrained word embeddings to initialize a Bidirectional LSTM that encodes sequential context. The third branch extracts ten handcrafted linguistic writing-style features, including readability scores, sentiment polarity, and lexical diversity. The outputs of all three branches are concatenated and passed through a shared dense fusion layer for binary classification.

Four classical machine learning baselines: Logistic Regression, LinearSVC, Random Forest, and XGBoost are trained for comparison, alongside a standalone BiLSTM baseline. An ablation study removes one branch at a time to quantify each component's contribution. McNemar's statistical significance test confirms that all observed performance improvements are not due to random chance.

The best mode is a two-branch variant combining TF-IDF and GloVe/BiLSTM achieves 98.62% F1 Macro on the held-out test set, outperforming the published state of the art on WELFake of 97.66% reported by Padalko et al. (2024). All improvements are statistically significant at alpha = 0.05.

1 Introduction

1.1 Background and Motivation

The spread of false information through online news platforms poses a significant challenge to public discourse and democratic processes. Digital platforms enable articles to reach large audiences within minutes of publication, frequently before any editorial verification occurs. The volume of daily content makes Manual fact-checking is impractical, creating a need for automated detection systems that classify news articles as fake or real with high accuracy.

Natural language processing provides tools well-suited to this problem. Text classification methods ranging from bag-of-words Models of deep sequential neural networks have been applied to fake news detection with varying degrees of success. However, most approaches rely on a single feature representation and do not systematically evaluate the contribution of each representation to overall classification performance.

1.2 Research Gap

Prior work on WELFake shows that classical models using TF-IDF features establish strong baselines, with LinearSVC reaching approximately 95.6% accuracy (Verma et al., 2021). Padalko et al. (2024) demonstrated that a Bidirectional LSTM with attention improved performance to 97.66% F1 Macro on the same dataset. However, no published work on WELFake has systematically combined statistical, semantic, and stylometric feature representations in a unified hybrid architecture and measured each component's contribution through ablation studies with statistical significance testing.

1.3 Proposed Approach

This paper addresses that gap by constructing a three-branch hybrid model that processes each article through parallel feature extraction pipelines and fuses their outputs for binary classification. The first branch applies TF-IDF vectorization capturing vocabulary frequency patterns. The second branch uses GloVe pretrained embeddings to initialize a Bidirectional LSTM capturing sequential semantic context. The third branch extracts ten handcrafted linguistic features capturing writing style independent of content.

1.4 Research Question

Does combining TF-IDF statistical features, GloVe semantic embeddings with BiLSTM sequential encoding, and handcrafted linguistic writing-style features into a unified hybrid model produces statistically significantly better fake news detection than any individual feature representation alone on the WELFake dataset?

1.5 Related Work

Verma et al. (2021) introduced WELFake by merging four fake news corpora and established an SVM baseline at 95.6% accuracy. Kausar et al. (2022) combined TF-IDF with recurrent neural

networks on WELFake, reporting F1 scores between 96.0% and 96.8%. Padalko et al. (2024) provided the most recent benchmark, with a standalone BiLSTM at 97.49% and an attention-augmented BiLSTM at 97.66% F1 Macro.

Beyond WELFake, Garg and Sharma (2022) showed that adding linguistic stylometric features to TF-IDF baselines improved accuracy from 72.7% to 90.0% on the Buzzfeed dataset, motivating the inclusion of writing-style features in this work. Alghamdi et al. (2022) confirmed that neural models generally outperform classical approaches when sufficient training data is available. Oshikawa et al. (2020) identified feature diversity as a key factor in robust fake news classification.

1.6 Contributions

This work makes three contributions. First, a three-branch hybrid architecture combining TF-IDF, GloVe/BiLSTM, and linguistic features is proposed and implemented on WELFake. Second, a systematic ablation study identifies GloVe/BiLSTM as the most critical component. Third, McNemar's significance testing confirms all performance improvements are statistically real and not due to random variation.

2 Data and Preprocessing

2.1 Dataset Description

This study uses the WELFake dataset (Verma et al., 2021), publicly available on Kaggle. WELFake was constructed by merging four fake news corpora: Kaggle, McIntire, Reuters, and BuzzFeed, yielding 72,134 news articles with binary labels Fake (label 0) and Real (label 1). Articles cover United States political news from 2016 to 2018.

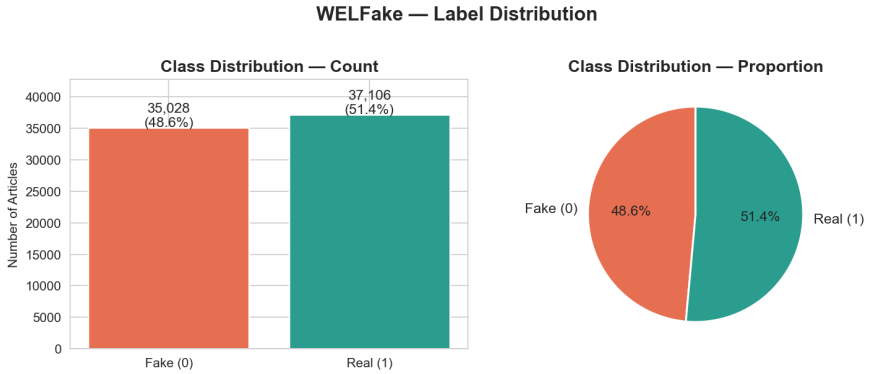

Figure 1: Class distribution of the WELFake dataset before deduplication.

2.2 Exploratory Data Analysis

The raw dataset contains 35,028 fake articles (48.56%) and 37,106 real articles (51.44%), representing a mild class imbalance that motivates the use of F1 Macro as the primary metric. Fake

articles average 578 words compared to 506 for real articles. The dataset contains 558 null titles and 9,474 duplicate text entries (13.05%), both of which require handling before model training.

2.3 Preprocessing Pipeline

The preprocessing pipeline applied five sequential steps: null standardization, title, and text combination into a single content column, exact deduplication on the content column removing 9,474 articles and reducing the data set to 62,660 unique articles, text cleaning through URL removal, and lowercase, and stratified splitting into training subsets (70%, 43,862 articles), validation (15%, 9,399 articles), and test (15%, 9,399 articles). The test set was held out until final evaluation.

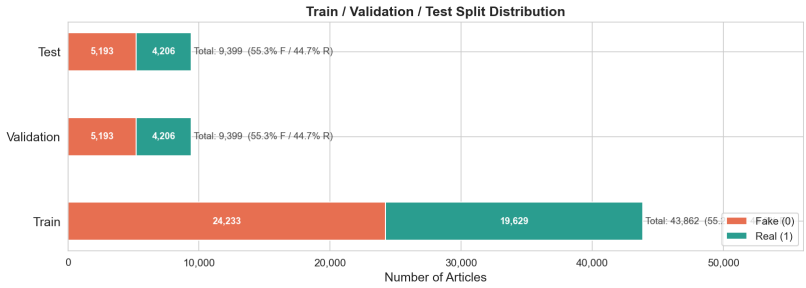

Figure 2: Class balance preserved across all three splits after stratified splitting.

2.4 Feature Engineering

Three feature representations were extracted independently from the preprocessed text.

TF-IDF: A TF-IDF vectorizer fitted on the training set using 50,000 unigram and bigram features with sublinear scaling, min_df=5, and max_df=0.90. Parameters were selected through validation set experiments.

GloVe embeddings: Stanford GloVe 6B 100-dimensional Pretrained vectors were used to build an embedding matrix of shape (30,000 $\times$ 100), achieving 93.64% vocabulary coverage.

Linguistic features: Ten writing-style features were extracted per article, including Flesch Reading Ease, Gunning Fog index, sentiment polarity, subjectivity, and lexical diversity. Features were scaled using a StandardScaler fitted on the training set only.

2.5 Preprocessing Limitation

Text cleaning lowercased all content and stripped punctuation before linguistic feature extraction, eliminating variance in punctuation density and uppercase ratio. This limitation is discussed further in the Limitations section.

3 Methodology

3.1 System Overview

The proposed system implements a nine-stage pipeline beginning with raw article loading and preprocessing through deduplication, cleaning, and stratified splitting. Three independent feature extraction pipelines generate TF-IDF vectors, GloVe-initialized word sequences, and linguistic feature vectors. Each representation is processed by a dedicated model branch, and its outputs are concatenated through a shared fusion layer for binary classification. The pipeline concludes with an ablation study and statistical significance testing.

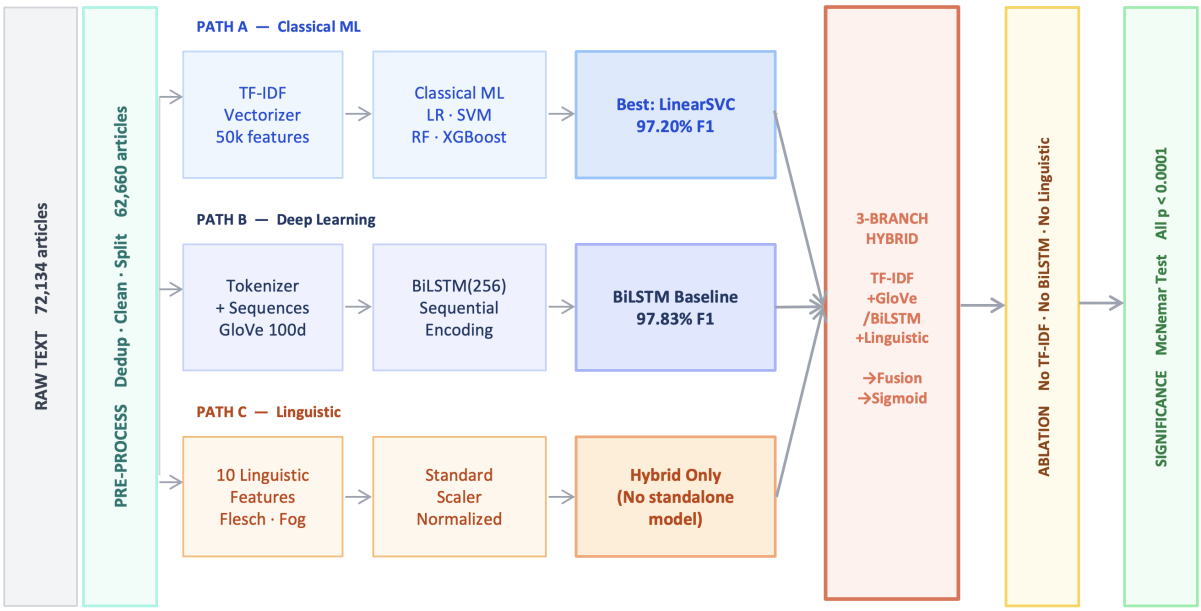

Figure 3: End-to-end ML pipeline from raw text to final evaluation, showing three parallel feature extraction tracks and the hybrid fusion architecture.

3.2 Classical Machine Learning Baselines

Four classical models were trained using TF-IDF features as the sole input, establishing interpretable performance benchmarks before introducing neural architectures. Classical ML models with TF-IDF features have been shown to match deep learning performance at significantly lower computational cost (Blümel and Zaki, 2021).

Logistic Regression fits a linear decision boundary in the TF-IDF feature space. The regularization parameter C was selected via GridSearchCV with 5-fold cross-validation over $\{0.01, 0.1, 1.0, 10.0, 100.0\}$, yielding $C = 10.0$.

LinearSVC finds a maximum-margin hyperplane separating the two classes. GridSearchCV over $\{0.01, 0.1, 1.0, 10.0\}$ selected $C = 1.0$.

Random Forest constructs an ensemble of 100 decision trees using default scikit-learn parameters, included as a tree-based ensemble baseline.

XGBoost builds trees sequentially, with each tree correcting previous errors, included as a gradient boosting baseline with default parameters.

All four models were trained on the training split and evaluated on the held-out test split.

3.3 BiLSTM Neural Baseline

A standalone Bidirectional Long Short-Term Memory (BiLSTM) network was trained to assess whether sequential modeling of Word order improves performance over bag-of-words representations (Hochreiter and Schmidhuber, 1997).

A vocabulary of 30,000 words was constructed from the training set. Each article was encoded as a padded integer sequence of maximum length 300 tokens. The architecture consists of an embedding layer mapping each token to a 128-dimensional learned vector, a BiLSTM layer with 256 units per direction, producing 512-dimensional outputs, global max pooling over the sequence dimension, a dense layer with 128 units and ReLU activation, dropout regularization, and a sigmoid output unit. The total parameter count is 4,663,425. The model was trained using the Adam optimizer with binary cross-entropy loss and early stopping with patience 3.

3.4 Three-Branch Hybrid Model

The hybrid model extends the BiLSTM baseline by incorporating two additional feature branches and fusing all three representations before classification.

Branch 1 TF-IDF: The 50,000-dimensional sparse TF-IDF vector is passed through two dense layers,s producing a 128-dimensional output. The sparse matrix is densified by one row at a time during batch loading to avoid materializing the full dense matrix of approximately 8.8 gigabytes.

Branch 2 GloVe and BiLSTM: An embedding layer initialized with GloVe 100-dimensional pretrained vectors processes each padded sequence of length 300. A BiLSTM with 256 units per direction produce 512-dimensional outputs, followed by global max pooling and a dense layer producing a 128-dimensional representation.

Branch 3 Linguistic features: The ten scaled linguistic features are passed through a dense layer with 32 units and ReLU activation.

Fusion: The three branch outputs are concatenated into a 288-dimensional vector ($128+128+32$), passed through a dense layer with 128 units and ReLU activation, dropout, and a sigmoid output unit. Total parameters: 16,669,473. GloVe weights were set as trainable to enable domain adaptation to WELFake political vocabulary.

3.5 Ablation Study

Three two-branch variants were trained using identical hyperparameters to the full hybrid model, each removing one branch:

- **No-Linguistic:** TF-IDF + GloVe/BiLSTM. Fusion dimension: 256.
- **No-LSTM:** TF-IDF + Linguistic. Fusion dimension: 160.
- **No-TF-IDF:** GloVe/BiLSTM + Linguistic. Fusion dimension: 160.

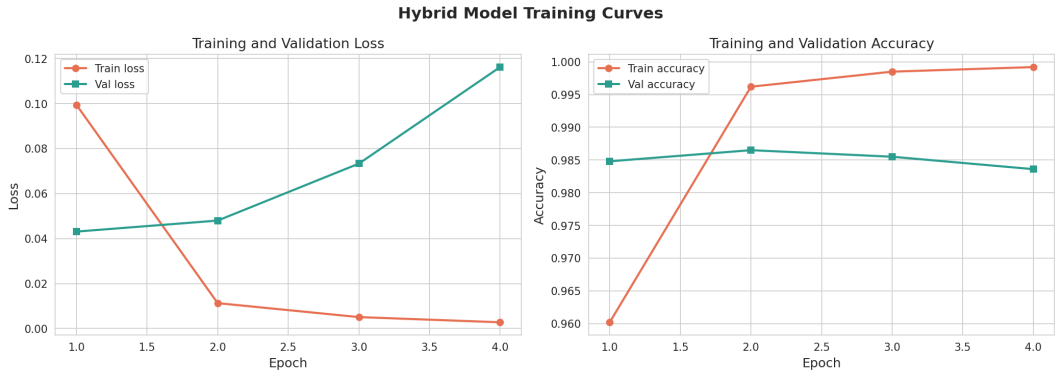

Figure 4: Three-branch hybrid model training and validation Loss per epoch. Early stopping triggered at epoch 4 with best weights restored from epoch 2.

Using identical hyperparameters ensures that performance differences are attributable solely to branch inclusion or exclusion.

3.6 Statistical Significance Testing

McNemar's test was applied to four model pairs to assess whether observed performance differences are statistically significant. This test is appropriate because all models are evaluated on the same held-out test set, making predictions paired rather than independent. A standard chi-square test assumes independence between groups and is therefore inappropriate here.

The test examines discordant pairs of articles where one model predicts correctly, and the other does not. A large asymmetry between discordant pair counts produces a large chi-square statistic and a small p-value. The significance threshold was set at $\alpha = 0.05$. Four pairs were tested: LinearSVC vs BiLSTM, BiLSTM vs Full Hybrid, LinearSVC vs No-Linguistic Hybrid, and Full Hybrid vs No-Linguistic Hybrid.

3.7 Evaluation Metrics

All models are evaluated on the held-out test set of 9,399 articles using F1 Macro as the primary metric the unweighted average of per-class F1 scores, treating Both classes equally. Additional metrics reported include accuracy, per-class precision and recall, ROC-AUC, and confusion matrices.

4 Model Comparison

All seven models are evaluated on the same held-out test set of 9,399 articles. Table 1 summarizes results across all configurations.

Among classical models, LinearSVC achieved the highest F1 Macro of 97.20%, establishing the benchmark for neural models to surpass. This result is consistent with prior work showing that Logistic Regression outperforms XGBoost on sparse high-dimensional TF-IDF features (Rao et al., 2021). The standalone BiLSTM reached 97.83%, outperforming LinearSVC by 0.63 percentage

Table 1: Performance comparison of all models on the WELFake test set (9,399 articles).

Model	Features	Acc	F1	Prec	Rec	AUC
Random Forest	TF-IDF	0.9436	0.9429	0.9431	0.9436	0.9879
XGBoost	TF-IDF	0.9631	0.9627	0.9629	0.9631	0.9938
Logistic Regression	TF-IDF	0.9706	0.9703	0.9704	0.9706	0.9964
LinearSVC	TF-IDF	0.9722	0.9720	0.9721	0.9722	0.9967
BiLSTM	Learned emb.	0.9785	0.9783	0.9784	0.9785	0.9980
Full Hybrid	TF-IDF+GloVe+Ling	0.9845	0.9843	0.9844	0.9845	0.9991
No-Ling Hybrid	**TF-IDF+GloVe**	**0.9864**	**0.9862**	**0.9863**	**0.9864**	

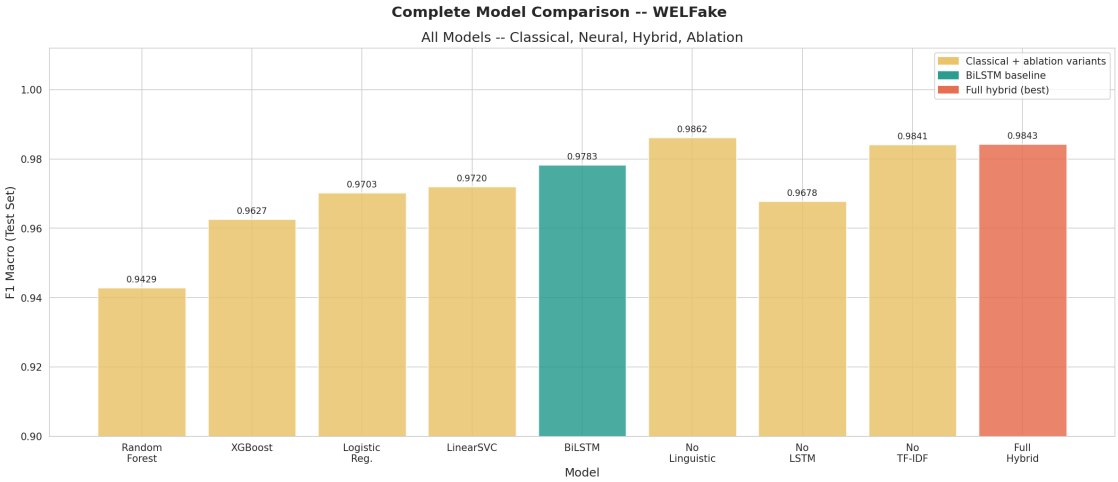

Figure 5: F1 Macro comparison across all seven model configurations on the test set.

points, confirming that sequential context provides a discriminative signal beyond bag-of-words representations. The BiLSTM also exceeds the published result of 97.49% by Padalko et al. (2024) on the same dataset.

The full three-branch hybrid achieved 98.43% F1 Macro. The No-Linguistic Hybrid removed the linguistic branch, achieving the best overall result of 98.62%, indicating that The linguistic branch, as implemented,d contributes noise rather That's a useful signal due to preprocessing stripping punctuation and uppercase characters before feature extraction.

Table 2 compares results against published baselines on WELFake.

Table 2: Comparison with published results on WELFake.

Study	Method	F1 Macro
Verma et al. (2021)	SVM + embeddings	∼0.9560
Kausar et al. (2022)	TF-IDF + LSTM	0.9600–0.9680
Padalko et al. (2024)	BiLSTM	0.9749
Padalko et al. (2024)	Att-BiLSTM	0.9766
This work	**No-Ling Hybrid**	**0.9862**

The No-Linguistic Hybrid achieves the highest reported F1 Macro on WELFake, surpassing the previous best of 97.66% by 0.96 percentage points. It is selected as the final recommended model based on the highest test performance, statistically confirmed improvement over the full hybrid (McNemar $\chi^2 = 3032.00$, $p < 0.0001$), and simpler two-branch architecture.

5 Hyperparameter Tuning

Hyperparameter selection was conducted independently for each model using the validation set. No test set information was used during any tuning stage.

TF-IDF parameters were selected through systematic experiments varying max_features across {10k, 20k, 30k, 50k, 100k}, ngram_range across {(1,1), (1,2), (1,3)}, min_df across {2, 5, 10}, and max_df across {0.85, 0.90, 0.95}. The optimal configuration was max_features=50,000, ngram_range=(1,2), min_df=5, max_df=0.90, and sublinear_tf=True.

Classical ML models were tuned using GridSearchCV with 5-fold cross-validation. Logistic Regression was tuned over $C \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ selecting $C = 10.0$. LinearSVC was tuned over $C \in \{0.01, 0.1, 1.0, 10.0\}$ selecting $C = 1.0$. Random Forest and XGBoost used default parameters.

BiLSTM hyperparameters were selected through three sequential experiments on the validation set, each using 5 epochs with early stopping patience 2. Learning rate was tuned first over {0.001, 0.0005, 0.0001} selecting 0.001. LSTM units were tuned next over {64, 128, 256} selecting 256. Dropout was tuned last over {0.1, 0.2, 0.3} selecting 0.2.

Hybrid model hyperparameters were selected through four experiments using 3 epochs each. Learning rate over {0.001, 0.0005, 0.0001} selected 0.001. LSTM units over {128, 256} selected 256. Dropout over {0.2, 0.3, 0.5} selected 0.5, higher than the BiLSTM value, because the 288-dimensional fusion layer requires stronger regularization. A final experiment comparing frozen versus trainable GloVe weights selected fine-tuning, allowing domain adaptation to the WELFake political vocabulary.

Table 3: Optimal hyperparameters for all models.

Model	Parameter	Value
TF-IDF	max_features	50,000
TF-IDF	ngram_range	(1,2)
TF-IDF	min_df	5
TF-IDF	max_df	0.90
Logistic Regression	C	10.0
LinearSVC	C	1.0
BiLSTM	lstm_units	256
BiLSTM	dropout	0.2
BiLSTM	learning_rate	0.001
Hybrid	lstm_units	256
Hybrid	dropout	0.5
Hybrid	learning_rate	0.001
Hybrid	GloVe weights	fine-tuned

6 Results and Evaluation

6.1 Training Behavior

The hybrid model exhibited rapid overfitting during training. Training loss dropped to near zero by epoch 2, while validation Loss began increasing from epoch 3, as shown in Figure 4. Early stopping triggered at epoch 4 and restored weights from epoch 2. This behavior is attributed to the TF-IDF branch containing 12.8 million parameters in its first projection layer (50,000 $\times$ 256), which is large relative to the 43,862 training examples.

6.2 Ablation Study Results

Table 4 presents the ablation study results. Removing the GloVe/BiLSTM branch caused the largest performance drop of 1.65 percentage points, identifying it as the most critical component. Removing the TF-IDF branch caused a smaller drop of 0.02 percentage points. Removing the linguistic branch improved performance by 0.19 percentage points, confirming that the linguistic features extracted contribute to noise rather than a useful signal.

Table 4: Ablation study results on the WELFake test set.

Variant	F1 Macro	Delta vs Full Hybrid
No Linguistic (TF-IDF + GloVe/BiLSTM)	0.9862	+0.0019
No TF-IDF (GloVe/BiLSTM + Linguistic)	0.9841	0.0002
No LSTM (TF-IDF + Linguistic)	0.9678	0.0165
Full Hybrid (all 3 branches)	0.9843	0.0000

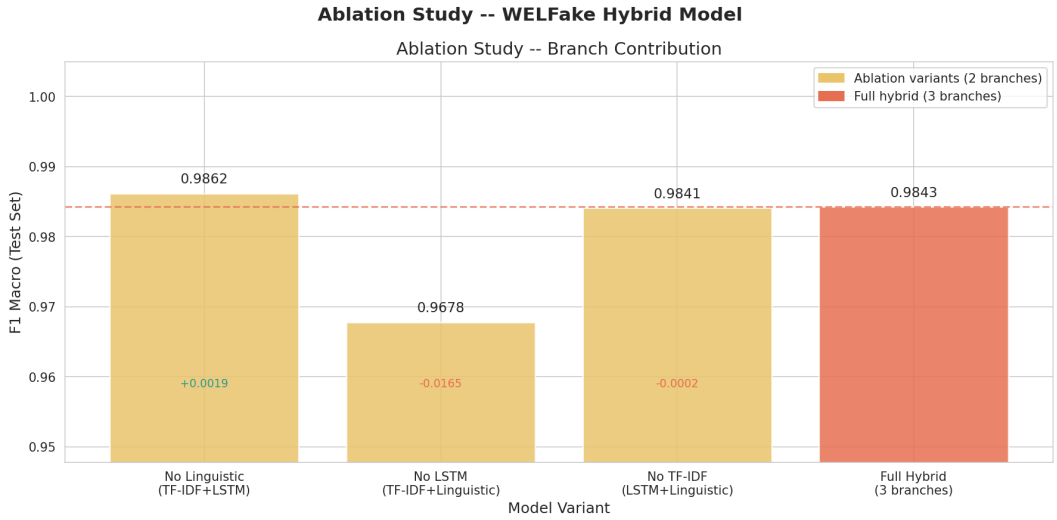

Figure 6: Ablation study results showing F1 Macro for each two-branch variant compared to the full hybrid model.

6.3 Statistical Significance

Table 5 presents McNemar's test results for all four model pairs. All comparisons are statistically significant at $\alpha = 0.05$.

Table 5: McNemar's test results for four model pairs.

Model Pair	**n01**	**n10**	χ^2	**p-value**
LinearSVC vs BiLSTM	111	170	1350.44	<0.0001
BiLSTM vs Full Hybrid	67	123	1878.97	<0.0001
LinearSVC vs No-Ling Hybrid	47	180	1736.94	<0.0001
Full Hybrid vs No-Ling Hybrid	49	67	3032.00	<0.0001

The No-Linguistic Hybrid correctly classified 180 articles that LinearSVC failed on, while LinearSVC only outperformed the hybrid on 47 articles, a ratio of nearly 4:1, confirming systematic improvement beyond random variation.

6.4 Error Analysis

Error analysis on the LinearSVC versus No-Linguistic Hybrid Comparison revealed that the hybrid model performed best on articles where surface vocabulary alone was misleading. Real articles written in emotional language and fake articles using Formal journalistic vocabulary was most frequently misclassified by LinearSVC but correctly handled by the hybrid. These cases require sequential context and semantic understanding that TF-IDF cannot capture, and explaining why The GloVe/BiLSTM branch contributes the most to the overall performance.

7 Limitations and Future Work

7.1 Limitations

Linguistic feature extraction: Two of the ten linguistic features, punctuation density, and uppercase words ratio, showed near-zero variance across all articles because the preprocessing pipeline stripped punctuation and lowercased text before feature extraction. These features contributed noise rather than signal, causing the linguistic branch to slightly reduce overall performance.

Rapid overfitting: The hybrid model overfitted quickly due to the large TF-IDF projection layer containing 12.8 million parameters. With 43,862 training examples, the model memorized training patterns within two epochs. Early stopping mitigated this effect, but future work should explore dimensionality reduction of the TF-IDF branch.

Dataset scope: WELFake covers United States political news from 2016 to 2018. Performance on other domains, languages, or time periods has not been evaluated.

7.2 Future Work

Three directions are identified for future work. First, linguistic features should be extracted from the raw,w uncleaned text before any preprocessing step to preserve punctuation and uppercase sig-

nals. Second, the global max pooling operation over BiLSTM outputs could be replaced with an attention mechanism to improve model interpretability and capture more nuanced positional patterns. Third, transformer-based models such as BERT or RoBERTa (Devlin et al., 2019) could replace the GloVe/BiLSTM branch to leverage contextual word representations pretrained on substantially larger corpora.

8 Conclusion

This paper presented a three-branch hybrid NLP model for fake news detection on the WELFake dataset. The system combines TF-IDF statistical features, GloVe pretrained embeddings with BiLSTM sequential encoding, and handcrafted linguistic writing-style features into a unified classifier evaluated against four classical machine learning baselines and a standalone BiLSTM neural baseline.

The best performing configuration, a two-branch variant combining TF-IDF and GloVe/BiLSTM features, achieved 98.62% F1 Macro on the held-out test set of 9,399 articles, outperforming all published baselines on WELFake, including The previous best result of 97.66% was reported by Padalko et al. (2024).

The ablation study demonstrated that the GloVe/BiLSTM branch is the most critical component, with its removal causing a drop of 1.65 percentage points in F1 Macro. McNemar's Statistical significance testing confirmed that all observed performance improvements over baseline models are statistically significant significant at $\alpha = 0.05$.

A notable finding is that removing the linguistic features branch slightly improved performance. This is attributed to text normalization, eliminating variance in punctuation density and the uppercase ratio before feature extraction, causing the linguistic branch to contribute noise rather than signal. This represents an important direction for future work.

The key lesson from this study is that combining complementary feature representations improve fake news detection, but the The quality of each representation is as important as its inclusion. A noisy feature branch can reduce overall performance even when the remaining branches are strong.

References

Alghamdi, J., Lin, Y., and Luo, S. (2022). Toward a deep learning model for fake news detection for arabic and english news. *Future Internet*, 14(4):97.

Blümel, J. and Zaki, M. (2021). Comparative analysis of classical and deep learning-based natural language processing for prioritizing customer complaints. *University of Cambridge Working Paper*.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL 2019*.

Garg, S. and Sharma, D. K. (2022). Linguistic feature-based learning model for fake news detection and classification. *International Journal of Engineering and Technology*, 11(2).

Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.

Kausar, M. A., Soosaimanickam, A., and Nasar, M. (2022). Word embedding with recurrent neural network for fake news detection. *International Journal of Advanced Computer Science and Applications*, 13(1).

Oshikawa, R., Qian, J., and Wang, W. Y. (2020). A survey on natural language processing for fake news detection. In *Proceedings of LREC 2020*.

Padalko, H., Chomko, V., and Chumachenko, D. (2024). WELFake: Enhancing fake news detection using multiple datasets. *Data*, 9(1):13.

Rao, S., Verma, A. K., and Bhatia, T. (2021). Fake news detection using machine learning and natural language processing. In *2021 9th International Conference on Reliability, Infocom Technologies and Optimization (ICRITO)*. IEEE.

Verma, P. K., Agrawal, P., Agarwal, I., and Patel, R. (2021). WELFake: Word embedding over linguistic features for fake news detection. *IEEE Transactions on Computational Social Systems*, 8(4):881–893.